

Section 12.3: The Dot Product

Vectors and the Geometry of Space

MTH 310
Calculus III

The Dot Product

We can **add** vectors and **scale** them. But can we "multiply" two vectors?

The **dot product** takes two vectors and gives a **scalar** (a number).

Dot Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Multiply corresponding components and add.

Computing Dot Products

Example 1

Compute each dot product.

$$\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle$$

$$\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle = 2(3) + 4(-1) = 6 - 4 = 2$$

$$\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle$$

$$\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle = -6 + 14 - 2 = 6$$

$$(\vec{i} + 2\vec{j} - 3\vec{k}) \cdot (2\vec{j} - \vec{k})$$

$$\langle 1, 2, -3 \rangle \cdot \langle 0, 2, -1 \rangle = 0 + 4 + 3 = 7$$

$$\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2 \text{ — the dot product of a vector with itself gives the **square of its magnitude**.}$$

Properties of the Dot Product

Properties

$$1. \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$2. \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} \quad (\text{commutative})$$

$$3. \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \quad (\text{distributive})$$

$$4. (c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$$

$$5. \vec{0} \cdot \vec{a} = 0$$

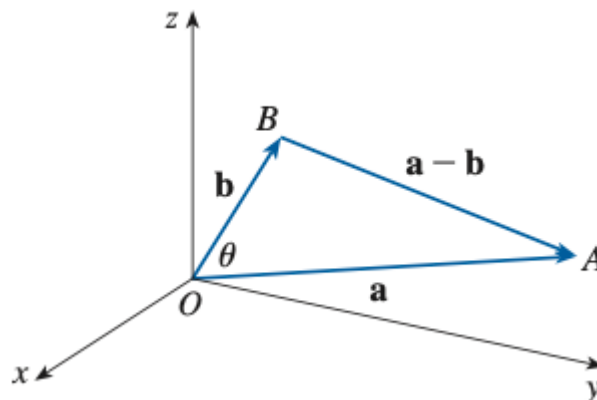
These follow directly from the component definition. Property 1 is especially useful — it connects the dot product to length.

The Geometric Meaning — Angles

Geometric Form of the Dot Product

If θ is the angle between $\vec{a}$ and $\vec{b}$ (with $0 \leq \theta \leq \pi$), then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

The diagram shows a triangle with vertices O, A, and B. Vector $\mathbf{a}$ is represented by a blue arrow from O to A. Vector $\mathbf{b}$ is represented by a red arrow from O to B. Vector $\mathbf{a} - \mathbf{b}$ is represented by a green arrow from A to B. The angle between vectors $\mathbf{a}$ and $\mathbf{b}$ at vertex O is labeled θ .

Example 2

Find the angle between $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$.

Find the angle between $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$.

$$|\vec{a}| = \sqrt{1 + 9 + 16} = \sqrt{26}, \text{ \texttt{ \hspace{0.5cm}}} |\vec{b}| = \sqrt{25 + 4 + 1} = \sqrt{30}$$

$$\theta = \arccos\left(\frac{5}{\sqrt{780}}\right) \approx 79.7^\circ$$

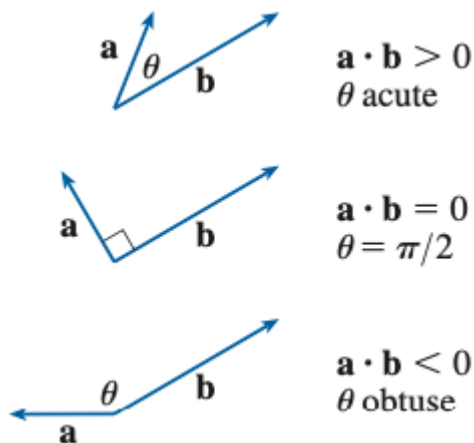
Orthogonal Vectors

Two vectors $\vec{a}$ and $\vec{b}$ are **orthogonal** (perpendicular) if and only if

$$\vec{a} \cdot \vec{b} = 0$$

Why? Because $\cos 90^\circ = 0$, so $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cdot 0 = 0$.

Sign of $\vec{a} \cdot \vec{b}$	Angle between vectors
$\vec{a} \cdot \vec{b} > 0$	Acute ($\theta < 90^\circ$)
$\vec{a} \cdot \vec{b} = 0$	Right angle ($\theta = 90^\circ$)
$\vec{a} \cdot \vec{b} < 0$	Obtuse ($\theta > 90^\circ$)



This is the **most important** use of the dot product — a quick test for perpendicularity!

Your Turn — Orthogonality Check

Example 3

Are $\vec{u} = \langle 2, 1, -1 \rangle$ and $\vec{v} = \langle 1, -1, 1 \rangle$ orthogonal?

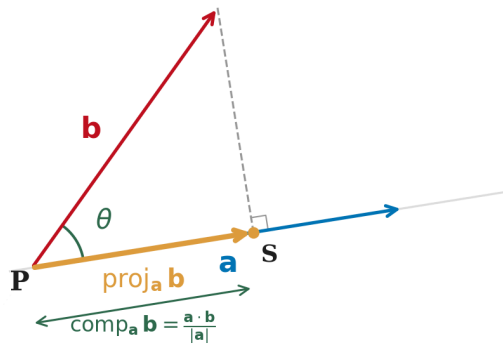
Compute u dot v and check if it equals zero.

$$\vec{u} \cdot \vec{v} = 2(1) + 1(-1) + (-1)(1) = 2 - 1 - 1 = 0$$

Yes, $\vec{u}$ and $\vec{v}$ are **orthogonal**!

Projections

How much of $\vec{b}$ points in the direction of $\vec{a}$?



Scalar and Vector Projections

Scalar projection (component of $\vec{b}$ onto $\vec{a}$):

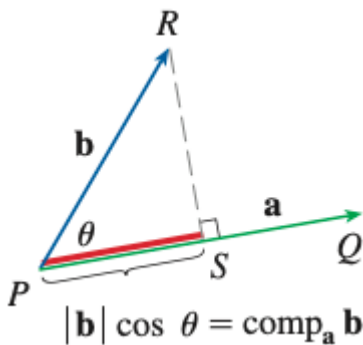
$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

This is a **number** — the signed length of the shadow.

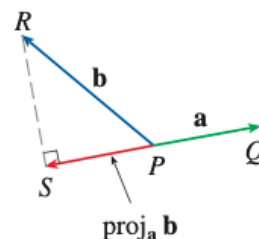
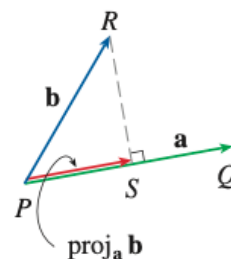
Vector projection of $\vec{b}$ onto $\vec{a}$:

$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

This is a **vector** — the actual shadow vector along $\vec{a}$.



comp = how far, \text{proj} = the actual vector



Example: Computing Projections

Example 4

Find the scalar and vector projections of $\vec{b} = \langle 12, 1, 2 \rangle$ onto $\vec{a} = \langle -1, 4, 8 \rangle$.

Compute a dot b , then $|a|$, then use the projection formulas.

$$\vec{a} \cdot \vec{b} = (-1)(12) + (4)(1) + (8)(2) = -12 + 4 + 16 = 8$$

$$|\vec{a}| = \sqrt{1 + 16 + 64} = \sqrt{81} = 9$$

$$\text{comp}_{\vec{a}} \vec{b} = \frac{8}{9}$$

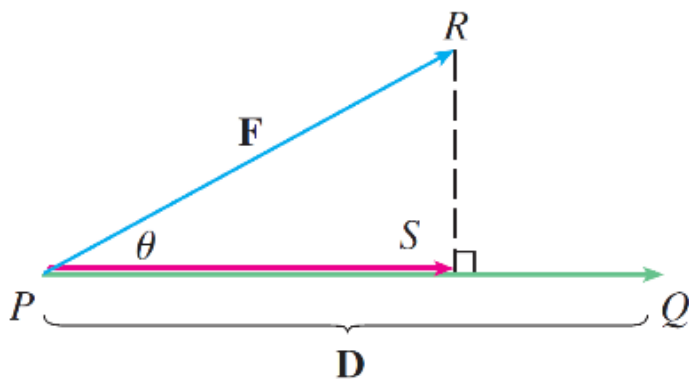
$$\text{proj}_{\vec{a}} \vec{b} = \frac{8}{81} \langle -1, 4, 8 \rangle = \left\langle -\frac{8}{81}, \frac{32}{81}, \frac{64}{81} \right\rangle$$

Work — A Physical Application

When force is aligned with motion: $W = |\vec{F}| \cdot |\vec{D}|$

When force is at an angle θ to the motion:

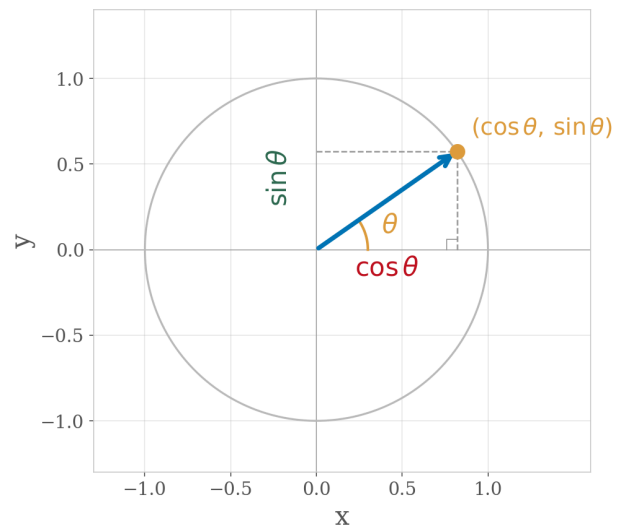
$$W = \text{comp}_{\vec{D}} \vec{F} \cdot |\vec{D}| = \vec{F} \cdot \vec{D}$$



Example 5

A wagon is pulled 100 m by a force of 70 N applied at 35° above horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.



$$\vec{F} = 70\langle \cos 35^\circ, \sin 35^\circ \rangle, \text{ \textit{ \textless \textless 100, 0 \textgreater \textgreater }}$$

$$W = \vec{F} \cdot \vec{D} = 70 \cos 35^\circ \cdot 100 + 70 \sin 35^\circ \cdot 0 = 7000 \cos 35^\circ \approx 5734 \text{ N} \cdot \text{m}$$

Your Turn — Work

Example 6

A box is pushed 5 m along the floor by a 40 N force applied at 20° below horizontal. Find the work done.

Set up the force and displacement vectors, then compute $F \cdot D$.

The horizontal component of the force does the work:

$$W = 40 \cos 20^\circ \cdot 5 = 200 \cos 20^\circ \approx 187.9 \text{ J}$$

Homework

Section 12.3: 5, 7, 11, 19, 23ab, 27, 39, 41, 45, 49, 57